



Koneru Lakshmaiah Education Foundation

(Category -1, Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

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Campus: Green Fields, Vaddeswaram - 522 302, Guntur District, Andhra Pradesh, INDIA.

Phone No. +91 8645 - 350 200; www.klef.ac.in; www.klef.edu.in; www.kluniversity.in

Admin Off: 29-36-38, Museum Road, Governorpet, Vijayawada - 520 002. Ph: +91 - 866 - 3500122, 2576129

Capstone Project Guidelines and Evaluation Rubrics

The Capstone Project is a culminating assignment undertaken by students in the final year of the programme, allowing them to apply the knowledge and skills gained over previous years of study. Students are expected to design, develop, and evaluate solutions to complex engineering problems using appropriate technology, tools, and engineering practice.

1. Programme Outcomes Addressed

- Think critically and solve challenging, open-ended problems by designing innovative solutions using acquired technical expertise.
- Apply engineering knowledge to real-world problems through innovative project development.
- Design and implement systems, components, or prototypes using modern tools, techniques, and interdisciplinary skills.
- Demonstrate effective teamwork, leadership, and project management abilities.
- Communicate technical information clearly and professionally through documentation and presentations.
- Adhere to professional and ethical standards, giving due consideration to societal, environmental, and sustainability aspects.

2. Procedure and Roles

1. Students are formed into batches of 3–5 members from the same specialization, working under the guidance of an assigned project guide.
2. Each section/class is assigned a dedicated Section In-Charge, responsible for monitoring the progress of all projects within that section.
3. Each department constitutes an Internal Review Panel comprising the Project Guide, a Mentor, and one or two additional faculty members with relevant domain expertise.
4. The evaluation process comprises three internal reviews (Review 1, 2, and 3) followed by one Final End-Semester External Review. Each internal review is assessed through three components: Continuous Evaluation, Guide Evaluation, and a Panel/Project Review — described in detail in Section 3.
5. Continuous Evaluation is conducted through in-session presentations delivered by students during scheduled project hours, at a minimum frequency of two presentations per month per batch.

Note: Faculty must upload marks to the LMS immediately after each session or review. Each batch is responsible for meeting its project guide at least twice a week and ensuring the guide records evaluation marks in the LMS accordingly. Failure to comply with these timelines may result in loss of continuous evaluation marks.

3. Assessment Structure and Weightage

The Capstone Project is evaluated out of 100 marks in total, split across the In-Semester (60 marks) and End-Semester (40 marks) components as shown below.

Component	Review 1	Review 2	Review 3	Final External Review	Total
Continuous Evaluation	5	5	5	—	15
Guide Evaluation	5	5	5	included below*	15
Panel / Project Review	10	10	10	—	30
Final End-Semester External Review	—	—	—	40	40
Total	20	20	20	40	100

**The Final External Review is scored out of 100 marks using its own rubric (Section 9), which itself includes a 20-mark Guide Evaluation criterion. This 100-mark score is then scaled down to 40 marks — see the scaling formulas below.*

3.1 How Component Scores Combine (Scaling Formulas)

Each review is assessed using rubrics scored out of 50 marks (Sections 6–8) or 100 marks (Section 9). These raw scores are then scaled down to fit the weightage shown above, as follows:

Scaling formula: *Continuous Evaluation (per review)* = $(\text{Average of all session scores in that review period} \div 50) \times 5$

Scaling formula: *Guide Evaluation (per review)* = $(\text{Guide's Individual Evaluation Sheet score} \div 50) \times 5$

Scaling formula: *Panel / Project Review (per review)* = $(\text{Panel Evaluation Rubric score} \div 50) \times 10$

Scaling formula: *Final External Review contribution* = $(\text{End-Semester External Review score} \div 100) \times 40$

These formulas allow every rubric to be filled out on its natural 0–50 or 0–100 scale during the review, while still rolling up consistently into the 100-mark overall total.

4. General Review Guidelines (Applicable to All Reviews)

Guide Presence is Mandatory: The internal project guide must be present at every review, as their evaluation and professional insight are integral to the process.

Panel Constitution: Each review panel must include the Project Guide and one or more Subject Experts / Review Committee Members.

Adherence to Schedule: Reviews must be conducted as per the department's academic calendar. Delayed reviews without prior approval may result in deduction of marks.

Pre-Submission of Documents: Students must submit all required documents (e.g., abstract, review report, slides) at least 2 days before the scheduled review date.

Attendance is Compulsory: All group members must be physically present. Absence without a valid, pre-approved reason will affect marks and may result in failure of the capstone project for that member.

Time Allocation: Each group is allotted 10–15 minutes to present, followed by 5–10 minutes for questions and discussion.

Academic Integrity: All content presented — including reports, code, and presentations — must be original. Plagiarism or misrepresentation of work will result in severe academic consequences, up to and including disqualification from the review. Departments are encouraged to run submitted reports through a plagiarism-detection tool prior to the review.

Post-Review Responsibilities

- Feedback must be recorded and shared with students after every review.
- Marks must be documented and signed by the guide and all panel members before submission.
- If a project is found incomplete or not meeting academic standards, a re-review must be scheduled within a timeframe set by the department.

4.1 Minimum Performance Requirements

To be filled in by the department in line with University academic regulations.

Minimum marks required in each individual review	<i>e.g., 40% of that review's total</i>
Minimum aggregate marks required across all reviews	<i>e.g., 50% overall</i>
Consequence of not meeting the minimum	<i>e.g., mandatory re-review, or referral to the department committee</i>

5. Continuous Evaluation

Continuous Evaluation is conducted batch-wise during scheduled project hours, at a minimum of two sessions per month per batch. Each session is scored using the rubric below; the average of all session scores within a review period is scaled to contribute 5 marks to that review (see Section 3.1).

S.No	Criterion	Max Marks	Marks Awarded	Remarks
1	Understanding of Project <i>Full marks: student can clearly explain project goals, current status, and their own individual contribution.</i>	10		
2	Quality of Presentation <i>Full marks: presentation is well-structured, visually clear, and appropriately paced.</i>	10		
3	Communication Skills <i>Full marks: ideas are conveyed clearly and confidently; questions are answered well.</i>	10		
4	Teamwork & Collaboration <i>Full marks: active, constructive participation and coordination with team members is evident.</i>	10		
5	Technical Progress <i>Full marks: demonstrable, on-track progress has been made since the previous session.</i>	10		
Total Marks		50		

Evaluations may be recorded per student or per team, with an optional remarks column used to note individual differences. Maintain a running project evaluation sheet for transparency and tracking.

6. Review 1 — Project Initiation & Planning

- Focus: understanding of the problem statement, literature survey, and planned methodology.
- Students should demonstrate clarity in objectives, a well-researched background and rationale, and initial planning covering team roles, tools selected, and timelines.
- The panel should evaluate feasibility and scope early, to help teams avoid major pivots later in the project.

Students must present:

- Problem statement and objectives
- Literature survey summary
- Proposed methodology and initial project plan (roles, tools, timeline)

Panel Evaluation Rubric — Review 1

S.No	Criterion	Max Marks	Marks Awarded	Remarks
1	Problem Statement & Objectives <i>Full marks: the problem is precisely defined, with SMART objectives that directly address it.</i>	10		
2	Background & Literature Review <i>Full marks: the survey covers sufficient recent, credible sources and clearly identifies the gaps the project will address.</i>	10		
3	Methodology & Planning <i>Full marks: methodology is well-justified and feasible; a realistic plan with milestones, roles, and tools is presented.</i>	10		
4	Stakeholder & User Needs Analysis <i>Full marks: target users/stakeholders are clearly identified and their needs are used to shape requirements.</i>	10		
5	Presentation & Communication <i>Full marks: the presentation is clear, well-organized, and confidently delivered; questions are answered accurately.</i>	10		
Total Marks		50		

Guide's Individual Evaluation Sheet — Review 1

The guide completes this sheet independently for each student, based on meeting attendance, contribution to planning/design/implementation, active participation, and communication with the guide. Marks may vary between students on the same project according to individual involvement.

S.No	Criterion	Max Marks	Student 1	Student 2	Student 3	Student 4	Student 5
	Registration Number	—					
1	Problem Statement & Objectives	10					
2	Background & Literature Review	10					
3	Methodology & Planning	10					
4	Stakeholder & User Needs Analysis	10					
5	Presentation & Communication	10					
Total		50					

Guide Signature: _____	Date: _____
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7. Review 2 — System Design & Partial Implementation

- Focus: design decisions, architecture diagrams, and the first stages of implementation.
- The panel should confirm integration of Agile practices where applicable and assess team collaboration.

Students must present:

- System Requirements Specification (SRS)
- ER / UML / Architecture diagrams
- Working prototype(s) or initial module(s)

Panel Evaluation Rubric — Review 2

S.No	Criterion	Max Marks	Marks Awarded	Remarks
1	System Requirements & Functional Specs <i>Full marks: requirements are complete, unambiguous, and traceable to the problem statement.</i>	10		
2	System Design & Architecture <i>Full marks: architecture/ER/UML diagrams are complete, consistent, and justify key design decisions.</i>	10		
3	Initial Implementation / Prototype <i>Full marks: a working prototype demonstrates core functionality aligned with the design.</i>	10		
4	Use of Tools / Technologies <i>Full marks: tools and technologies are appropriate, correctly applied, and justified.</i>	10		
5	Teamwork & Agile Practice <i>Full marks: there is clear evidence of iterative development, sprint planning, and effective team collaboration.</i>	10		
Total Marks		50		

Guide's Individual Evaluation Sheet — Review 2

The guide completes this sheet independently for each student, based on meeting attendance, contribution to planning/design/implementation, active participation, and communication with the guide. Marks may vary between students on the same project according to individual involvement.

S.No	Criterion	Max Marks	Student 1	Student 2	Student 3	Student 4	Student 5
	Registration Number	—					
1	System Requirements & Functional Specs	10					
2	System Design & Architecture	10					
3	Initial Implementation / Prototype	10					
4	Use of Tools / Technologies	10					
5	Teamwork & Agile Practice	10					
Total		50					

Guide Signature: _____	Date: _____
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8. Review 3 — Final Implementation & Evaluation

- This is the most critical internal review, validating full functionality of the system, final testing results and performance metrics, and use-case scenarios demonstrating real-world applicability.
- Final documentation should be complete or near-final at this stage.
- Soft skills such as communication, reflection, and professionalism should also be assessed.

Students must present:

- Fully functional system demonstration
- Testing results and performance metrics
- Near-final project documentation

Panel Evaluation Rubric — Review 3

S.No	Criterion	Max Marks	Marks Awarded	Remarks
1	Completed System Functionality <i>Full marks: all planned features are implemented and functioning correctly and reliably.</i>	20		
2	Testing & Performance Evaluation <i>Full marks: comprehensive test cases are documented with clear results and performance metrics.</i>	10		
3	Real-world Use Cases / Impact <i>Full marks: the demonstration clearly shows practical applicability and value to intended users.</i>	10		
4	Reflection, Learning & Future Scope <i>Full marks: reflection on challenges and learnings is thoughtful, with a credible plan for future work.</i>	10		
Total Marks		50		

Guide's Individual Evaluation Sheet — Review 3

The guide completes this sheet independently for each student, based on meeting attendance, contribution to planning/design/implementation, active participation, and communication with the guide. Marks may vary between students on the same project according to individual involvement.

S.No	Criterion	Max Marks	Student 1	Student 2	Student 3	Student 4	Student 5
	Registration Number	—					
1	Completed System Functionality	20					
2	Testing & Performance Evaluation	10					
3	Real-world Use Cases / Impact	10					
4	Reflection, Learning & Future Scope	10					
Total		50					

Guide Signature: _____	Date: _____
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9. Final End-Semester External Review

Pre-Review Preparations

Guide Presence is Mandatory: the final review must not be conducted without the internal project guide's presence.

Document Submission (2–3 days before the review):

- Final project report (spiral/hardbound, signed by the guide)
- Presentation slides, as per department format
- User manual / supporting documents, if applicable
- Project demonstration link / code repository, if applicable

Project Display Setup: ensure laptops are functional, code is ready for live demonstration, and any required software/hardware is set up in advance.

Review Process

- Each team is allotted 20–25 minutes in total: 10–15 minutes for presentation and 5–10 minutes for Q&A and demonstration.
- The presentation must cover: problem statement, objectives, methodology, architecture/design, live demonstration, and outcomes with future scope.
- Grading must reflect technical depth, execution quality, learning outcomes, and practical applicability.

Post-Review Responsibilities

- Marks awarded by the examiner must be recorded and signed by all panel members.
- Constructive feedback should be given to each team, verbally or in writing.
- Final hardbound reports (with signatures) must be submitted for departmental archiving.

Evaluation Rubric — Final End-Semester External Review

S.No	Criterion	Max Marks	Marks Awarded	Remarks
1	Problem Definition & Objectives <i>Full marks: the problem and objectives are precisely stated and well-motivated.</i>	10		
2	Literature Review & Technical Background <i>Full marks: the work demonstrates thorough grounding in relevant, current literature and technology.</i>	10		
3	System Design & Architecture <i>Full marks: the design is complete, sound, and clearly documented.</i>	10		
4	Implementation & Functionality <i>Full marks: the system is fully implemented and functions as intended.</i>	10		
5	Testing, Validation & QA <i>Full marks: testing is rigorous, with clear evidence of validation and quality assurance.</i>	5		
6	Innovation / Use of Advanced Tools & Techniques <i>Full marks: the project demonstrates originality or effective use of advanced tools/techniques.</i>	10		
7	Team Collaboration & Project Management <i>Full marks: there is clear evidence of effective planning, coordination, and role distribution.</i>	5		
8	Report Quality & Documentation <i>Full marks: the report is complete, well-organized, and professionally presented.</i>	10		
9	Viva-Voce & Presentation Skills <i>Full marks: responses are confident and clear, with accurate articulation of the work.</i>	10		
10	Guide Evaluation <i>Full marks: reflects the guide's overall assessment of the student's/team's engagement, effort, and contribution across the project.</i>	20		
Total Marks		100		

Guide Signature: _____	Date: _____
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10. Score Consolidation Sheet

To be completed by the Section In-Charge after all reviews are complete, using the scaling formulas in Section 3.1. One sheet per student.

Student Name	
Registration Number	
Project Title	

Component	Raw Score	Scaled Score	Max
Review 1 — Continuous Evaluation	___ / 50	___	5
Review 1 — Guide Evaluation	___ / 50	___	5
Review 1 — Panel Review	___ / 50	___	10
Review 2 — Continuous Evaluation	___ / 50	___	5
Review 2 — Guide Evaluation	___ / 50	___	5
Review 2 — Panel Review	___ / 50	___	10
Review 3 — Continuous Evaluation	___ / 50	___	5
Review 3 — Guide Evaluation	___ / 50	___	5
Review 3 — Panel Review	___ / 50	___	10
Final End-Semester External Review	___ / 100	___	40
Overall Total		___	100

11. Document Submission Checklist

A consolidated checklist of documents expected at each stage of the Capstone Project.

Stage	Documents Required
Review 1	Problem statement, literature survey summary, initial project plan
Review 2	System Requirements Specification, architecture/ER/UML diagrams, working prototype
Review 3	Fully functional system, test reports and performance metrics, near-final documentation
Final External Review	Final hardbound report (signed), presentation slides, user manual (if any), code repository / demo link (if any)